BT402 FINAL TERM PAPER

M.C.Q for Final terms:

1. The starter cultures produce lactic acid, resulting in a pH decrease in the milk and promoting curd synthesis

2. Types of fermentation 2

3. Fermentation is the conversion of carbohydrates (plant foods) to alcohols and carbondioxide
4. In 1885 the French microbiologist Louis Pasteur and Emile roux developed the first vaccine against **Rabies**

5. Live, attenuated vaccines contain a version of the living microbe that has been weakened in the lab so it can't cause disease

•6. Toxoid vaccines are used when a bacterial toxin is the main cause of illness.
7. Conjugate vaccines developed for bacteria having polysaccharide coating.

8. An adjuvant is a substance that is formulated as part of a vaccine to enhance its ability to induce protection against infection

• 9. Antimicrobial products kill or slow the spread of microorganisms

10. Triclosan, an antimicrobial compound, has been found to promote oral health and aid in the fight against gingivitis when added to toothpaste.

11. The microorganisms are said to be chemolithotrophic ("rock-eating");

12. A **biofuel** is defined as any fuel whose energy is obtained through a process of biological carbon fixation

13. First-generation biofuels are made from sugar, starch, vegetable oil, or animal fats using conventional technology.

14. Cynobacteria relationship with plants Symbiotic

15. Bio-fertilizers add nutrients through the natural processes of nitrogen fixation, solubilizing phosphorus,

• 16. . Biopesticide is a formulation made from naturally occurring substances that controls pests by non toxic mechanisms and in eco-friendly manner.

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• 18. Composting is a natural process that turns organic material into a dark rich substance.

• 19. **Grasscycling** refers to an aerobic (requires air) method of handling grass clippings by leaving them on the lawn

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- 20. Prolonged use of certain antibacterial can decrease the number of gut flora which may have a negative impact on health
- 21.Ring worm disease treated by Antifungal
- 22. Large areas of genetically modified (GM) crops of soybeans, maize, cotton, and canola have been successfully grown in the Western Hemisphere.
- 23.if a. genes comes from a different species this is called horizontal gene transfer
- 24.The common type of GMOs plant is Alfalfa
- 25.Microorganisms came into being on the earth over a period of about 1.2-2.5 billion years

26. Microorganisms are used in the production of fermented foods and beverages Yeast, Penicillum lactobacillus all

- 27.GMOs may also cause a threat to the environment

28. Penicillin is made from the filamentous fungi

29. The fungi invade the organic matter in soil first and are then followed by the bacteria

30. Shear sensitivity is belongs to physiological factor

31. GMOs do not enough pesticides because the crop itself is dangerous to some insects

- 32.Yeast is responsible for the fermentation process which produce alcohol in wine

33. Chemical conservation of sugar into ethanol producing alcoholic drinks such as wine beer and cider

Vaccines 2

A vaccine is a biological preparation that improves immunity to a particular disease. A vaccine typically contains an agent that resembles a disease-causing microorganism and is often made from weakened or killed forms of the microbe.

Biochemical pesticides 2

They are naturally occurring substance to control pest by non-toxic mechanisms. Biochemical pesticides include substances as insect sex pheromones that interfere with mating that attract insect pest to traps.

First generation Biofuels 2

First-generation biofuels are made from sugar, starch, vegetable oil, or animal fats using conventional technology. These are generally produced from grains high in sugar or starch fermented into bioethanol.

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What are DNA vaccines 3

DNA vaccines take immunization to a new technological level. These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe's genetic material.

Example: Influenza vaccine.

Pharmaceutical products 3

Many pharmaceutical drugs are also produced by microbes e.g. **Cyclosporin A** that is used as an immunosuppressive agent in organ-transplant patients is produced by the fungus *Trichoderma polysporum*.

Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol-lowering agents. It acts by competitively inhibiting the enzyme responsible for the synthesis of **cholesterol**.

Second generation Biofuels 3

Second-generation biofuels are produced from non-food crops, such as cellulosic biofuels and waste biomass (stalks of wheat and corn, and wood). Common second-generation biofuels include vegetable oils, biodiesel, bioalcohols, biogas, solid biofuels, and syngas.

Research continues on second-generation biofuels including biohydrogen, biomethanol, DMF, Bio-DME, Fischer-Tropsch diesel, biohydrogen diesel, mixed alcohols and wood diesel.

What are Antimicrobial pesticides? 5

Antimicrobial pesticides help to slow the growth of human pathogenic microorganisms and are used in food processing plants, dairies, breweries, poultry houses and other animal feeding operations, hospitals and medical and dental clinics and offices, municipal drinking water and water treatment facilities, swimming pools and spas, schools, day-care centers, public access facilities and homes

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Interactions between plants and microbes 5

1. Many microbes are fond in nature and help plants to grow

2. Rhizobacteria found in the soil fixate nitrogen which is required to grow many crops

Plant-microbe interactions describe a broad range of scientific studies concerning. how microbes interact with plants at the molecular biology and molecular genetics level.

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• Plants and microbes can have a variety of interactions including

• Symbiotic

• Associative

• Pathogenic,

• All of which impact plant productivity, stress tolerance and disease resistance.

SYMBIOTIC ASSOCIATIONS WITH

CYANOBACTERIA

• Cyanbobacteria is broadly distributed in nature and form symbiotic relationships with many different organisms.

• Cyanobacteria enhance their survival by establishing an association with a biological partner.

• Azolla: is an aquatic fern that contains bilobed leaves attached to a stem and is found floating in freshwater.

• The cyanobacteria are found in a cavity between the ventral and dorsal epidermal layers of the leaf.

Associative relation of microbes and plants

ROOT ASSOCIATIONS

• The enzyme system for nitrogen fixation is found only in the prokaryotes, and in the case of symbiotic nitrogen fixation in plants,

• There is considerable specificity between the legume symbiont and bacteria for the stable association.

• The plant provides the carbon and energy source for the bacteria to grow, while the bacteria fix nitrogen with the production of amino acids for plant growth.

PATHOGENIC INTERACTION

Bacterial Pathogens

• The bacteria-producing diseases in plants generally display a number of hydrolytic enzymes for penetration of the plant surface, and growth in the plant is generally intercellular between the parenchyma cells.

• As the disease progresses in the plant, bacterial distribution may become systemic as a result of migration into the vascular tissue.

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• In some cases, the bacteria release pectinase that hydrolyzes the plant cell walls and the plant collapses.

How microbes enhance and improve the oil recovery? 5

One of the major concerns facing the oil industry today is the recovery of the large percentage of oil remaining unrecovered in mature and in nearly depleted oil fields.

• Microbes can influence and improve the oil recovery process by:

• Generating gases that increase reservoir pressure and decrease oil viscosity

• Generating acids that dissolve rock, thus improving absolute permeability

• Reducing permeability in channels

• Producing bio-surfactants that decrease interfacial tension

• Reducing oil viscosity by degrading long-chain saturated hydrocarbons

Role of microbes in petroleum industry. 10

Petroleum-based products are the major source of energy for industry and daily life.

Leaks and accidental spills occur regularly during the exploration, production, refining, transport, and storage of petroleum and petroleum products.

• Release of hydrocarbons into the environment whether accidentally or due to human activities is a main cause of water and soil pollution

• Soil contamination with hydrocarbons causes extensive damage of local system since accumulation of pollutants in animals and plant tissue may cause death or mutations.

Microbial processes play an important role in industry, the protection of the natural environment.

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